Precept 8 Outline – Key

ECO 202 TAs

November 12, 2026

1 Weekly Check In

1. Any questions about the class, comments?

2 Precept practice problems

1. Suppose independent customer purchase amounts have population mean 42 dollars and population standard deviation 18 dollars. The population distribution is right-skewed. Let $\bar{X}_n$ be the mean purchase amount in an independent sample of n customers.

   (a) Calculate $E[\bar{X}_{64}]$ and $SD(\bar{X}_{64})$ exactly under the stated model.

   (b) Use the CLT to approximate $P(\bar{X}_{64} > 45)$. Show the standardized boundary and interpret the probability in repeated-sampling terms.

   (c) Repeat the calculation for n = 256 and explain why the upper-tail probability shrinks even though the population mean and threshold remain unchanged.

   (d) State one simulation that could assess finite-sample accuracy for both sample sizes if the complete population model were available. Explain why right skew makes the check useful and what the simulation would not establish about real customers.

   Solution. (a) $\mathbb{E}[\bar{X}_{64}] = 42$ and $\text{SD}(\bar{X}_{64}) = 18/\sqrt{64} = 2.25$. (b) The standardized boundary is $(45-42)/2.25 = 1.33$, so $\mathbb{P}(\bar{X}_{64} > 45) \approx 1 - \Phi(1.33) \approx 0.092$. Thus about 9.2% of repeated samples of 64 would have a mean above 45 under the model. (c) For $n = 256$, $\text{SD}(\bar{X}) = 18/16 = 1.125$ and the boundary is $(45-42)/1.125 = 2.67$, giving a tail probability of about 0.0038. The larger sample makes the sampling distribution more concentrated around 42. (d) Repeatedly simulate independent samples of sizes 64 and 256 from the complete population model, compute each sample mean, and compare the simulated fractions above 45 with the CLT values. Skew makes finite-sample approximation error worth checking; the simulation checks the assumed model, not whether that model describes real customers.

2. A 100-order sample has mean 12.60 minutes and sample standard deviation 2.50 minutes. Use the large-sample estimated standard error $2.50/\sqrt{100} = 0.25$ minutes and the approximation $Z \sim N(0,1)$ under $H_0 : \mu = 12$. You may use $P(N(0,1) \geq 2.40) = 0.0082$.

   (a) Calculate the standardized statistic and interpret its sign and magnitude.

   (b) Sketch or describe the two reference-distribution tails that count under $H_a : \mu \neq 12$.

   (c) Calculate the two-sided p-value.

   (d) Interpret that p-value as a probability under the null and procedure; do not assign a probability to the null hypothesis.

   (e) Apply the two-sided rule at $\alpha = 0.05$ and $\alpha = 0.01$. Use reject or fail-to-reject language.

   Solution. (a) $z = (12.60 - 12)/0.25 = 2.40$: the estimate is 2.4 estimated standard errors above the null value. (b) Count both $Z \geq 2.40$ and $Z \leq -2.40$. (c) $p = 2(0.0082) = 0.0164$. (d) If H_0 and the reference procedure were correct, the chance of a statistic at least this far from zero in either direction would be about 1.64%. (e) Reject at $\alpha = 0.05$; fail to reject at $\alpha = 0.01$.

3. A randomized outreach study estimates an average intention-to-treat effect δ on an application-completion rate, measured in percentage points. For this planning exercise, treat the estimator's reference model as

$$\hat{\delta} \sim N(\delta, 2^2)$$

Test $H_0 : \delta = 0$ against $H_a : \delta \neq 0$ at $\alpha = 0.05$ using $Z = \hat{\delta}/2$ and the rule $|Z| > 1.96$. The reference standard error 2 is treated as fixed across the alternatives considered.

(a) Describe a Type I error in the outreach setting.

(b) Describe a Type II error at the specific alternative $\delta = 3$ percentage points.

(c) Translate the rejection rule into critical values for $\hat{\delta}$ in percentage points.

(d) Under $\delta = 3$, show that $Z \sim N(1.5, 1)$ and calculate power using $P(N(0,1) > 0.46) = 0.3228$ and $P(N(0,1) < -3.46) = 0.0003$.

(e) Calculate the Type II error probability at $\delta = 3$.

(f) Suppose four times as many units under an otherwise comparable independent design halves the reference standard error to 1. Under $\delta = 3$, use $P(N(0,1) > -1.04) = 0.8508$ and a negligible opposite tail to approximate power. Explain the change.

(g) With design and alternative fixed, explain how lowering α affects Type I error and power.

Solution. (a) Rejecting H_0 and concluding outreach has an effect when $\delta = 0$. (b) Failing to reject when the true effect is 3 percentage points. (c) Reject when $|\hat{\delta}| > 1.96(2) = 3.92$. (d) Since $Z = \hat{\delta}/2$, under $\delta = 3$ it has mean $3/2 = 1.5$ and standard deviation 1. Power is $\mathbb{P}(Z > 1.96) + \mathbb{P}(Z < -1.96) = 0.3228 + 0.0003 = 0.3231$. (e) Type II error probability is $1 - 0.3231 = 0.6769$. (f) With standard error 1, the standardized mean is 3 and power is approximately 0.8508; the smaller standard error separates the alternative from zero more clearly. (g) Lowering α reduces Type I error but moves the critical values outward and therefore lowers power.

3 Extra practice problems

1. Suppose (Y, X) has the joint distribution

$Y \backslash X$	0	2
-1	0.00	0.20
1	0.50	0.30

and let (X_i, Y_i), $i = 1, 2, \ldots$, be independent pairs, each distributed like (X, Y).

(a) Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

(b) Use the CLT to approximate

$$\mathbb{P}\left(\sum_{i=1}^{100} X_i > 111\right).$$

Solution. (a) The marginal distribution has $\mathbb{P}(X = 0) = \mathbb{P}(X = 2) = 0.5$, so $\mathbb{E}[X] = 1$ and $\mathrm{Var}(X) = 1$. (b) By the CLT, $\bar{X}_{100} \dot{\sim} N(1, 0.1)$. Thus

$$\mathbb{P}\left(\sum X_i > 111\right) = \mathbb{P}(\bar{X}_{100} > 1.11) \approx \mathbb{P}\left(Z > \frac{1.11 - 1}{0.1}\right) = \mathbb{P}(Z > 1.10) \approx 0.136.$$

2. Suppose X has distribution

x	1	2	3
$\mathbb{P}(X = x)$	0.3	0.2	0.5

and $X_1, \ldots, X_{100}$ are independent and distributed like X.

(a) Find $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathrm{Var}(X)$.

(b) Use the CLT to approximate

$$\mathbb{P}\left(\frac{1}{100}\sum_{i=1}^{100} X_i > 2.3\right).$$

Solution. (a) $\mu = 1(0.3) + 2(0.2) + 3(0.5) = 2.2$, and $\sigma^2 = 0.3(1-2.2)^2 + 0.2(2-2.2)^2 + 0.5(3-2.2)^2 = 0.76$. (b) The standard error is $\sqrt{0.76}/10 \approx 0.0872$. Hence

$$\mathbb{P}(\bar{X}_{100} > 2.3) \approx \mathbb{P}\left(Z > \frac{2.3 - 2.2}{0.0872}\right) = \mathbb{P}(Z > 1.15) \approx 0.125.$$

3. A study tests the null hypothesis that 70% of all U.S. companies grant their executives stock options. In a simple random sample of 420 companies, 281 grant options. Using the large-sample estimated standard error, find the p-value of an appropriate two-sided hypothesis test. State the hypotheses and your decision at the 5% level.

Solution. $H_0 : p = 0.70$ versus $H_a : p \neq 0.70$. Here $\hat{p} = 281/420 \approx 0.6690$ and

$$\widehat{\mathrm{SE}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{420}} \approx 0.0230, \qquad z = \frac{0.6690 - 0.70}{0.0230} \approx -1.35.$$

Thus $p \approx 2\Phi(-1.35) \approx 0.18$, so fail to reject H_0 at the 5% level.

4. Let p_1 and p_2 be the fractions of individuals with a characteristic in two populations. Two reasonably large independent samples yield an estimated difference $\hat{p}_2 - \hat{p}_1 = 0.15$ with estimated standard error 0.10. Find the approximate p-value of the two-sided test of $H_0 : p_1 = p_2$, and state the decision at the 5% level.

Solution. $z = 0.15/0.10 = 1.50$, so the two-sided p-value is $2\mathbb{P}(Z \geq 1.50) \approx 0.1336$. Since $0.1336 > 0.05$, fail to reject H_0.

5. We observe two independent random variables $X \sim N(\mu, 2)$ and $Y \sim N(\mu, 5)$, where the second argument is the standard deviation. We test $H_0 : \mu = 0$ against $H_a : \mu > 0$. Define

$$\hat{\mu}_1 = \frac{X+Y}{2}, \qquad \hat{\mu}_2 = 0.6X + 0.4Y.$$

(a) Find the critical value for the 5% level test that rejects for large values of $\hat{\mu}_1$.

(b) Consider instead the test that rejects when $\hat{\mu}_2 > 1.2$. Find its power when $\mu = 1$.

Solution. (a) Under H_0, $\hat{\mu}_1 \sim N(0, \sqrt{29}/2)$. The critical value is $1.645(\sqrt{29}/2) \approx 4.43$. (b) When $\mu = 1$, $\hat{\mu}_2 \sim N(1, \sqrt{0.6^2(2^2) + 0.4^2(5^2)}) = N(1, \sqrt{5.44})$. Therefore

$$\text{Power} = \mathbb{P}(\hat{\mu}_2 > 1.2) = \mathbb{P}\left(Z > \frac{1.2 - 1}{\sqrt{5.44}}\right) \approx \mathbb{P}(Z > 0.086) \approx 0.466.$$

6. We observe independent random variables

$$X \sim N(\mu, 1), \qquad Y \sim N(2\mu + 1, 3),$$

where the second argument is the standard deviation. We test $H_0 : \mu = 0$ against $H_a : \mu < 0$.

(a) A test rejects when $X + Y < c$. Choose c so that the Type I error probability is 5%.

(b) A different test rejects if and only if both $X < -1$ and $Y < 0$. Find its power when $\mu = -1$.

Solution. (a) Under H_0, $X + Y \sim N(1, \sqrt{10})$. Thus $c = 1 - 1.645\sqrt{10} \approx -4.20$. (b) At $\mu = -1$, $X \sim N(-1, 1)$ and $Y \sim N(-1, 3)$. Independence gives

$$\text{Power} = \mathbb{P}(X < -1)\mathbb{P}(Y < 0) = 0.5\,\Phi(1/3) \approx 0.315.$$